

Special topics in logic and security II *

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1. A code C consists of the following four words:

$$(0, 0, 0, 0, 0, 0, 0, 0); (1, 1, 0, 0, 1, 1, 0, 0); (0, 0, 1, 1, 0, 0, 1, 1); (1, 1, 1, 1, 1, 1, 1, 1).$$

- (a) Show that this code is linear over $\mathbb{F}_2$. Find its parameters $[n, k, d]$.
- (b) Find a generating matrix G for C .
- (c) Find a check-matrix H for C .
- (d) Show that $C \subset C^\perp$.
- (e) Find the minimal distance of $C^\perp$.

2. Let C be the following code over the field $\mathbb{F}_{11}$.

$$C = \{(f(1), f(2), \dots, f(7)) \mid f(x) = ax^2 + bx + c \in \mathbb{F}_{11}[x]\}.$$

- (a) Show that one can always correct one mistake.
 - (b) Correct the word $(6, 3, 4, 9, 7, 9, 8)$.
3. Let $S \subset \mathbb{C}$ be the set of complex numbers of radius $\frac{1}{\sqrt{2}}$. Let $B \subset \mathbb{C}^2$ be the set of all qubits. Show that $S \times S \subset B$. Show also that $B \neq S \times S$.
4. On a table at CERN there are two independent globes. The red globe has the quantum state $r = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ while the blue globe has the quantum state $b = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$. We interpret by $|1\rangle$ that a mutation happens in the globe.
- (a) How probable is to separately observe mutations in these globes?
 - (b) We consider the composite system of the two globes. Compute its quantum state.
 - (c) How probable is to observe a mutation in the composite system?
5. Let H_n be a complex Hilbert space. A unitary function $U : H_n \otimes H_n \rightarrow H_n \otimes H_n$ such that for all states $|x\rangle, |y\rangle \in H_n$ one has:

$$U(|x\rangle|y\rangle) = |x\rangle|x\rangle$$

is called a general quantum copy machine. Show that there is no general quantum copy machine.

6. OPTIONAL QUESTION

Consider the triangles ABC with $A = 90^\circ$ and $AB + BC = \frac{3}{2}$. Which of them has the maximal area?

*Master, 4-th semester.